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The anterior cingulate cortex directs exploration of alternative strategies

Highlights

- Foraging task exposes ongoing strategy switching with two-component computations
- ACC perturbation at corresponding two time points has opposing effect on switching
- The two computations map onto distinct interacting sub-circuits within the ACC

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In brief

Tervo et al. combine a foraging task in rats with ensemble recordings and modern circuit perturbation tools to map two-component computations of ongoing strategy arbitration—whether to persist with the default strategy and whether to commit to an encountered alternative—onto distinct interacting subcircuits within the ACC.

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The anterior cingulate cortex directs exploration of alternative strategies

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SUMMARY

The ability to adjust one's behavioral strategy in complex environments is at the core of cognition. Doing so efficiently requires monitoring the reliability of the ongoing strategy and, when appropriate, switching away from it to evaluate alternatives. Studies in humans and non-human primates have uncovered signals in the anterior cingulate cortex (ACC) that reflect the pressure to switch away from the ongoing strategy, whereas other ACC signals relate to the pursuit of alternatives. However, whether these signals underlie computations that actually underpin strategy switching or merely reflect tracking of related variables remains unclear. Here we provide causal evidence that the rodent ACC actively arbitrates between persisting with the ongoing behavioral strategy and temporarily switching away to re-evaluate alternatives. Furthermore, by individually perturbing distinct output pathways, we establish that the two associated computations—determining whether to switch strategy and committing to the pursuit of a specific alternative—are segregated in the ACC microcircuitry.

INTRODUCTION

Making good choices in uncertain and dynamic situations is a core challenge of intelligent behavior. The requisite flexible action planning, thought to be implemented by the prefrontal cortex (Miller and Cohen, 2001), effectively maps environmental and internal states onto behavioral strategies and prospectively searches through this mapping to select a suitable strategy. However, in open-ended natural settings, this mapping is potentially infinite, rendering a search for the optimal strategy computationally intractable. An emerging view is that our brains meet this challenge by employing approximations of optimal solutions, arbitrating among a small set of concurrently monitored behavioral strategies before pursuing formation of new ones (Cleland et al., 2001; Donoso et al., 2014). Defining the circuit logic that permits the brain to balance pursuit of an ongoing strategy with exploration of alternatives is therefore a central issue in neural control of behavioral adaptation.

One prefrontal area, the anterior cingulate cortex (ACC), has long been viewed as key to behavioral adaptation in complex settings. The ACC may be especially critical when the task state or temporal context, inferred through experience and deliberation, calls for overriding the ongoing behavioral strategy (Shenhav et al., 2017). Indeed, functional imaging in humans has uncovered rich contextual representations of evaluative signals in the ACC (Shenhav et al., 2016) as well as ACC activation during departure from the ongoing strategy to re-explore previously learned alternatives (Blanchard and Gershman, 2018; Cleland et al., 2001). Furthermore, findings in humans (Behrens et al., 2007; Kolling et al., 2012, 2014; McGuire et al., 2014; O'Reilly et al., 2013; Schuck et al., 2015), non-human primates (Blanchard and Hayden, 2014; Hayden et al., 2009; Procyk et al., 2000), and rodents (Karlsson et al., 2012; Ma et al., 2016; Powell and Redish, 2016) have implicated the ACC in keeping track of alternative strategies and learning environmental statistics that signal the need to switch strategies (Kolling et al., 2014; Quilodran et al., 2008). However, whether this area plays an active role in strategy switching or merely monitors environmental context and behavioral performance remains unclear.

In this study, we directly probe the ACC's role in ongoing evaluation of strategies through targeted optogenetic perturbations at distinct key time points during the decision-making process. Taking advantage of a rat behavioral task design that segregates the period of default strategy re-evaluation and the point of commitment to pursuit of an alternative, we demonstrate that perturbing ACC activity during these two task epochs leads to opposing effects on persistence with the default strategy. Combined with an electrophysiological signature of strategy switching we uncover in targeted recordings of ACC ensemble activity, these findings demonstrate that the ACC plays an active dual

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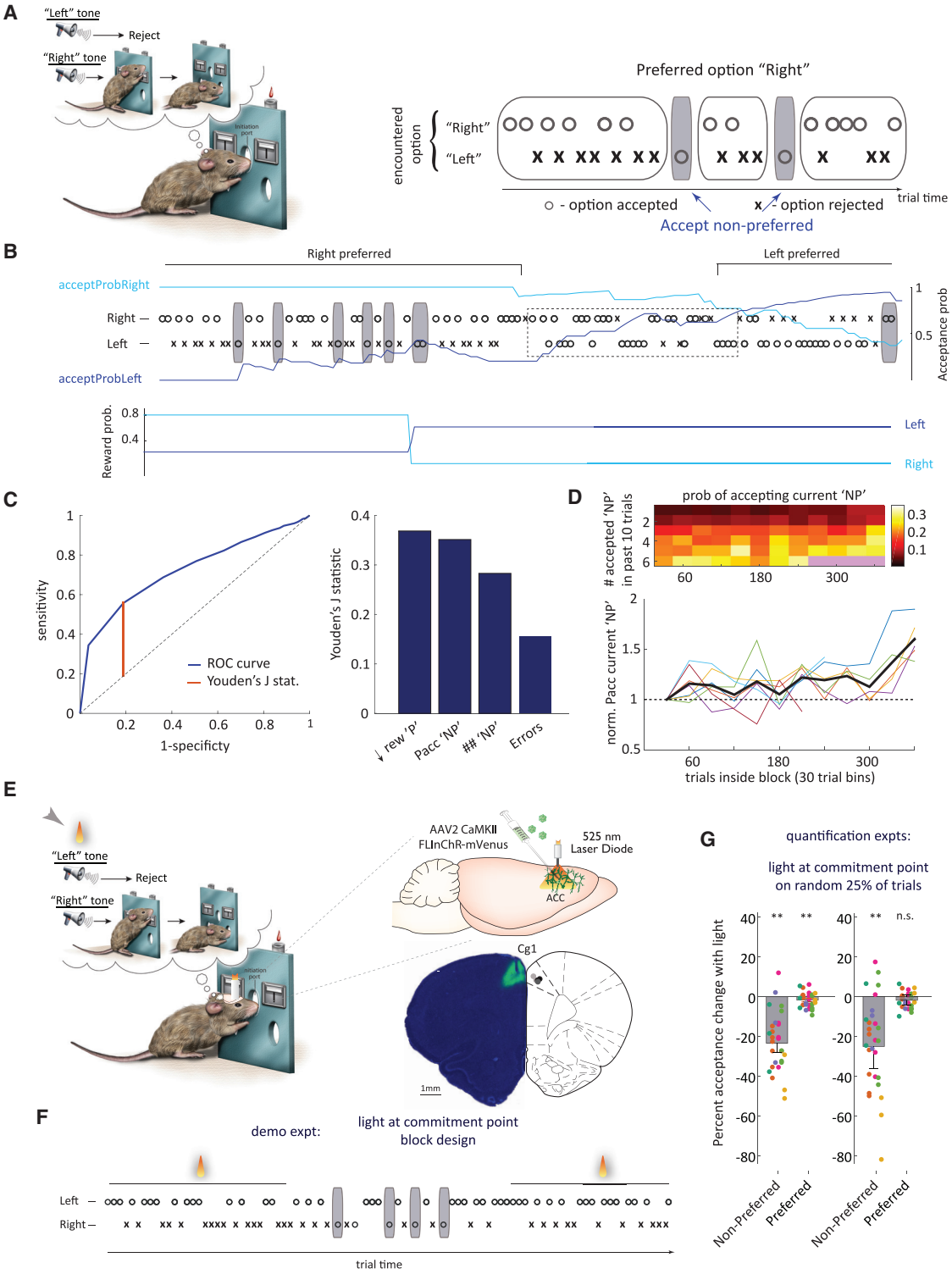

Figure 1. ACC perturbation at the commitment point reduces strategy switching in a foraging context

(A) Left panel: task schematic. Sequential, randomly iterated, tone-cued presentation of two behavioral options leads to a series of accept/reject decisions. Right panel: a block-wise structure promotes local resolve to pursue one option (here, right) at the expense of the other (here, left). This default strategy is occasionally interrupted by decisions to accept the non-preferred option (gray shading).

(B) Top panel: sample behavioral trace around a block transition. Note a marked preference at the beginning (for the right choice option) and end (for the left choice option) and sporadic deviations from the preferred option (gray shading). A dashed box highlights a transient period of resampling following inferred

(legend continued on next page)

role in evaluating the strategy space: in determining whether to persist with the ongoing strategy, and—when strategy switching is deemed appropriate—in guiding exploration of alternatives. Using virally mediated access to individual ACC output pathways, we then reveal that this dual role relies on anatomically distinct circuits. Overall, our findings argue that an opponent interaction within the ACC microcircuitry determines the balance between persisting with an ongoing strategy and switching to evaluate alternatives.

RESULTS

A behavioral framework that exposes ongoing strategy switching

We trained rats in a behavioral paradigm (Karlsson et al., 2012) that captures the essential structure of sequential decision-making during foraging (Barack and Platt, 2017; Stephens et al., 2004) by presenting a series of iterated accept-or-reject choices for two tone-cued options, each paired with distinct reward probabilities (Figure 1A; STAR Methods). An emerging view based on foraging theory argues that economic choices involve sequential evaluation of single options, each comprising a decision about whether to accept/reject that option relative to the set of possible alternatives (Hayden, 2018; Kacelnik et al., 2011; Pearson et al., 2014). Accordingly, in our framework, rats are presented—while at a central initiation port—with sequential options (left or right) signaled by one of two auditory tones that, with a certain probability, provide a liquid sugar reward after an imposed travel time, implemented as a requirement to perform several lever presses (Figure 1A). Animals can accept the option by pressing the corresponding lever for a possible reward or reject it by re-initiating at the central port. Because this temporally structured sequential process more closely resembles natural patch/prey encounters during foraging than experimental settings built around a forced choice between simultaneously presented alternatives, it may more easily expose core decision-making processes.

In noisy environments subject to abrupt changes, animals balance pursuit of a currently favored behavioral strategy with sporadic re-evaluation of alternatives (Costa and Averbeck, 2020; Costa et al., 2019; Ebitz et al., 2018; Ellwood et al., 2017; Gallistel et al., 2001; Tchernichovski et al., 1998; Wilson et al., 2014), and our expert animals often displayed a strong commitment to a single (preferred) option during blocks of stable reward probabilities (Figures 1B and S1A) by accepting nearly all trials of one type and rejecting most of the other. Notably, the pursuit of the preferred option was occasionally interrupted by sporadic acceptances of the non-preferred one (Figures 1A and 1B, gray rectangles). Because accept/reject decisions are inherently asymmetric, with acceptance/pursuit representing an active process that likely triggers monitoring and adjustment and rejection representing passive continuation of the search for the favored option (Hayden, 2018), the sporadic acceptance of the non-preferred option reflects transient switching of behavioral strategy. Such ongoing strategy switches are distinct from abrupt transitions to exploration in response to environmental state changes (in our study, short bouts of resampling of both options following unsignaled block transitions; Figure 1B, dashed box) that have been explored recently in the context of possible ACC engagement by a complementary set of perturbation studies in rodents (Akam et al., 2021; Ellwood et al., 2017). Indeed, in contrast to the environmental change-triggered undirected exploratory bouts, the ongoing transient strategy switches away from a determined pursuit of the preferred option continued deep into blocks of stable reward probabilities (Figure S1B) even when the difference in the payoff between the two options was large (Figure S1C). The latter observation argues that our behavioral framework exposes a form of ongoing strategy arbitration that is not guided simply by the local fluctuations in reward but is likely influenced by more complex task and temporal contexts of a type that could engage the ACC.

We next sought to quantify the relationship between local task context, or more global temporal context, and the animal's choice. We used a receiver operating characteristic analysis (ROC) to examine how well the occurrence of particular behavioral features predicted animals' choices when rats are offered the non-preferred option. The features we examined fell into three main categories: parameters associated with an individual

environmental changes. Solid blue and cyan lines represent independent measures of running acceptance probability for the left and right options, respectively, used to identify preferred and non-preferred trials (STAR Methods). Bottom panel: block structure. Blue line, reward probability associated with the right option; cyan line, reward probability associated with the left option.

(C) Left: example ROC space that demonstrates the calculation of Youden's J statistic, a metric that captures the best dichotomization of data by a feature. Blue line, the ROC curve for a local downturn in reward associated with the preferred option; dashed line, random classification; orange line, maximum of Youden's J statistic. Right: maximum Youden's J statistic for three behavioral features that display the strongest discriminability: a recent decrease in reward associated with the preferred option (rew P), recent acceptance of the non-preferred option (Pacc NP), and the number of consecutive presentations of the non-preferred option (## NP). For comparison, Youden's statistic was also calculated for one of the less predictive features, the local error rate (Error).

(D) Top: prevalence of accept decisions for the non-preferred option as a function of the number of trials inside a block of stable probabilities (x axis, values estimated for 30 trial bins) and of the number of acceptances of non-preferred options within the past 10 trials (y axis). Black/white denote no/frequent acceptance of the non-preferred option; purple denotes insufficient data. Bottom: The same data, normalized to the initial acceptance rate for non-preferred options. Each line corresponds to different parameterizations of the number of accept decisions in the previous 10 trials. The black line represents the mean.

(E) Left: schematic of perturbation experiment. Light delivery was restricted to tone presentation (gray arrow), when the rat commits to an accept/reject action. Top right: schematic of the viral approach for opsin delivery. Bottom right: opsin expression. Circles indicate fiber placement.

(F) Accept/reject behavior in a perturbation experiment. Note greater acceptance of non-preferred options (gray shading) during no-light blocks versus light blocks.

(G) Illumination-dependent fractional change in acceptance of the non-preferred and preferred options throughout the session (left) or late into blocks of stable probabilities (right). Colors denote different rats. Bars represent average effects and SEM across within-animal means, n = 6 animals. Error bars represent SEM. n.s., not significant. **$p < 0.01$.

See also Figures S1 and S2.

trial leading up to the choice to be explained (at lags of 1–10 trials), running averages of different types of trials in the recent past, and the number of trials since the last occurrence of a particular type of trial (STAR Methods). Three behavioral features contributed prominently to animals' decisions to accept the non-preferred option: (1) recent (within the last 2–3 trials) reward omission associated with the preferred option, (2) recent (within the last 10–15 trials) acceptance of the non-preferred option (consistent with these decisions occurring in bouts), and (3) the run length of consecutive non-preferred option encounters (Figure 1C). The acceptance of non-preferred options in bouts suggests that a local downturn in experienced reward (lack of reward associated with the preferred option on a given encounter) is not the only factor driving strategy switching. Indeed, decisions to persist after such local downturns were not obviously driven by the preferred option being otherwise rewarded more or less in the recent past (Figure S1D; no significant difference in more than half of the sessions in the distributions of reward history between unrewarded preferred trials that were followed by the decision to persist or switch; Kolmogorov-Smirnov test). Furthermore, the strong relationship of local task-related features with acceptances of the non-preferred option (Figure 1C, right) argues that they are unlikely to be random, unintended failures to choose the correct behavioral response (that is, a response aligned with the ongoing default strategy).

We also investigated whether more global features—specifically, the temporal context—affect the animals' decisions to accept or reject the non-preferred option. We measured residuals from the local-feature prediction model and consistently found a gradual increase in acceptance of non-preferred options as a function of the number of trials within a block (Figures 1D and S1E). This observation suggests that rats may anticipate blocks nearing their end, possibly reflecting their ability to estimate environmental volatility (Behrens et al., 2007; Nassar et al., 2010). These analyses reveal that rats did not simply follow fluctuations in local reward but integrated task and temporal context when choosing their strategy.

A temporally structured decision-making task allows perturbation at specific time points with different computational relevance

The consideration that economic choice might reflect a series of sequential accept/reject decisions (Hayden, 2018; Kacelnik et al., 2011; Pearson et al., 2014) implies that two independent computations might underpin a switch from pursuit of the preferred option to explore alternatives: a decision to "try something else" and a temporally segregated decision to commit to a specific, subsequently encountered alternative. Therefore, separate circuits might contribute to strategy switching at these distinct time points—a possibility that our task design permits us to evaluate. To explore the role of the ACC in strategy arbitration, we therefore optogenetically perturbed its activity at two corresponding points during a trial: (1) during tone presentation, when the animal learns of encountered option's identity and commits to an action (accept/reject) dictated by his chosen strategy, and (2) immediately after delivery of feedback at the preferred trial's end, when the default strategy's success is likely evaluated and when the animal decides to persist with that default strategy or explore an alternative.

We ensured spatial selectivity of our perturbation by expressing a recently developed, robust light-dependent silencer, FLInChR (Brown et al., 2018), locally in the ACC (focusing specifically on region 24b within Cg1; Vogt and Paxinos, 2014; Figure 1E; STAR Methods) and by using modest light intensities (~1–2 mW). FLInChR-expressing rats performed 700–1,500 iterated accept/reject trials, a subset of which were paired with light delivery through bilaterally implanted optical fibers, with the duration of the light stimulus restricted to the period of tone presentation (Figure 1E). This design permitted us to compare behavior for light-on versus light-off trials within individual sessions, making the analysis robust to individual differences in the overall incidence of accept decisions for non-preferred trials, which varied across sessions. To simplify the analysis, we only included periods where there was a strongly preferred option (STAR Methods; such periods constituted about 75% of the dataset; Figure S1A).

ACC perturbation at the time of option encounter preferentially decreases acceptance of the non-preferred option

Light pulses during tone presentation led to a marked reduction in acceptance of the non-preferred option, with over 2-fold suppression observed in some sessions (Figures 1F and 1G; ratio of accept decisions for light-on versus light-off trials, 0.77 ± 0.05 across within-animal means, $p < 0.005$, Wilcoxon rank-sum test, $n = 6$ animals; see Figure S2A for the effect on absolute acceptance probability). This effect was equally strong when we excluded the transition periods following block changes (Figure 1G; ratio of accept decisions for light-on versus light-off trials, 0.75 ± 0.09, $p < 0.005$, Wilcoxon rank-sum test, $n = 6$ animals). Shuffle analysis also confirmed that the effect of ACC perturbation on acceptance of the non-preferred choice option did not arise by chance (Figure S2B) and was consistent across experimental animals (Figure S2C). In contrast, light application had a minor (<2%) effect on acceptance of preferred options (Figures 1F and 1G, ratio of accept decisions for light-on versus light-off trials 0.98 ± 0.01 across within-animal means; Figures S2A and S2B) that was not consistent across individual animals (Figure S2D). ACC perturbation thus preferentially decreases acceptance of an option that is currently not the preferred one.

The session-to-session variability in the effect of ACC perturbation on decision-making likely arose because of variability of the perturbation itself—expression efficiency, light delivery efficiency, and variation in the resulting dynamics within the local ACC circuitry—and not other experimental factors. The observed behavioral effects also did not result from a light-dependent increase in occasional (<5%) erroneous attempts to accept the option not indicated by the tone (ratio of errors for light-on versus light-off trials, 0.88 ± 0.07; error trials were not analyzed further and are omitted from the examples shown), nor did they stem from an inability to process tones or from an aversion to a specific side of the behavioral box because within-session changes in reward probabilities ensured that unaffected preferred and affected non-preferred trials comprised both choice options (left trials, paired with tone 1, and right trials, paired with tone 2). More parsimoniously, the observed

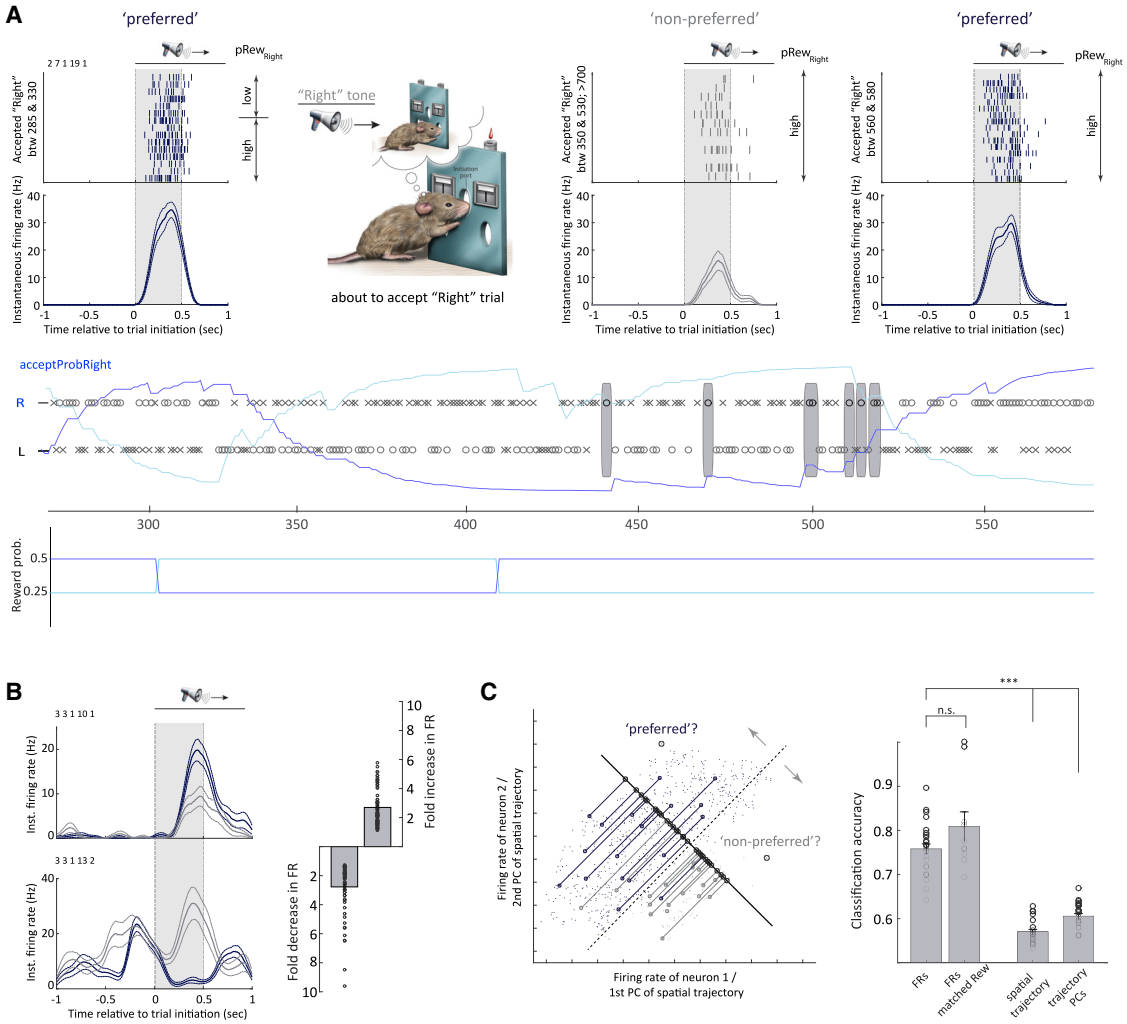

Figure 2. ACC ensemble activity reflects strategy switches

(A) Top: Activity for example cell around the point of commitment to the accept decision for the right option, which was first the default strategy, then part of the resurgent alternative, and back to default. The box indicates the 500 ms window used for the decoding analysis in (C). Center: corresponding behavior. Gray rectangles indicate when the right option was accepted despite being non-preferred. Bottom: block structure.

(B) Left: simultaneously recorded cells that show opposite modulation during the default strategy (blue) and resurgent alternative (gray). Right: summary of fold modulation for all cells that show differential activity depending on strategy. FR, firing rate.

(C) Left: schematic of linear classification. Test data are classified by their projection onto the hyperplane that best separates data from the training set. Right: classification accuracy for single sessions (points) and average (bars). FRs: accuracy for trials in default and resurgent groups; n = 4 animals, n = 25 sessions. FRs, matched reward: accuracy for trials subsampled from the two groups to match reward contingencies; n = 4 animals, n = 10 sessions with block structure confined to probability reversals. Spatial trajectory: accuracy for trials in the two groups based on the associated spatial trajectory, approximated as a series of x, y positions, binned at 100 ms, and represented as points in a 10-dimensional space; n = 4 animals, n = 25 sessions. Trajectory PCs: accuracy for trials in the two groups based on the three main principal components of the spatial trajectories; n = 4 animals, n = 25 sessions. Bars show SEM; ***$p < 0.001$.

behavioral change likely stems from the context (non-preferred trial) in which the accept/reject choice is made.

To eliminate the possibility that the perturbation effect arises because the ACC merely provides a non-specific excitatory drive to a hypothetical key downstream area in which this computation occurs (Otchy et al., 2015), we next used targeted extracellular recordings to verify that the ACC encodes signals relevant to the decision to accept a non-preferred option. We reasoned that, if the ACC is selectively engaged during the decision to accept a non-preferred as opposed to the preferred option, then the neural representation associated with a specific action (e.g., accepted right) likely differs, depending on the strategy-related context (preferred or non-preferred) (Figure 2A). Indeed, a substantial fraction of units recorded in ACC exhibited a marked difference in activity between the two contexts, specifically during presentation of the relevant tone, with some units exhibiting a significant increase and others a decrease in activity (significant modulation in 42% of all neurons active during trial initiation; Figure 2B). The patterns of ACC ensemble activity associated with acceptance of an option between trials when

the option was preferred or non-preferred were sufficiently different to permit training of a linear classifier (Figure 2C; classification accuracy, 75.2% ± 1.2%, n = 4 animals, n = 25 sessions), suggesting that this difference is distinct enough to be informative to downstream networks.

Trajectory information is represented prominently in ACC ensemble activity (Remondes and Wilson, 2013); thus, the comparison of ACC ensemble activity at the commitment point between strategy-related contexts is confounded by the fact that our animals would have often approached the initiation port using different trajectories originating at spatially distinct locations. To verify that the observed modulation of ACC ensemble activity does not simply reflect a difference in the kinematics of actions executed as a part of the preferred versus non-preferred choice, we repeated the linear classification analysis, now focusing on spatial trajectories. The classifier performed significantly worse when trained on the entire trajectory or on its most variable component (Figure 2C), suggesting that modulation of ACC activity cannot be explained by gross motor confounds.

We also evaluated whether the observed activity modulation merely reflects different reward contingencies. For this analysis, one would ideally compare ensemble activity for trials in the preferred or non-preferred groups, subsampled to match local reward history. However, we do not know how individual animals integrate local reward history, especially over randomly interleaved presentations of the two options. We thus focused on a subset of sessions where the probabilities of reward assigned to each block had a simpler structure; reward probabilities were confined to simple reversals between two values, permitting classification of trials into high or low reward categories (Figure 2A; STAR Methods). Because switches in reward blocks were uncued, animals typically continued choosing their previously preferred option for some time, despite it being rewarded less than the alternative (e.g., in Figure 2A, the right side reverts to being the lower-rewarded one at ~trial 300, but the behavior adapts around trial 350, leaving half of the accepted right trials in the first "default" block in the low reward category). Trials associated with acceptance of a specific option in the preferred and non-preferred groups could then be subsampled to match in the reward category. If the observed modulation of ensemble dynamics were determined by reward, then we would expect a decoder of the animal's preference to fail when trained and tested only on reward category-matched subsets. In contrast, we found that decoders did no worse when trained on reward category-matched datasets than on the non-reward matched datasets (including the entire dataset; Figure 2C; classification accuracy, 76.7% ± 1.2%, n = 3 animals, n = 10 sessions). Thus, more parsimoniously, the observed modulation stems from the specific strategy's context (preferred versus non-preferred) in which acceptance of a specific option was made. Combined with the perturbation experiments, these observations argue that the ACC actively directs resampling of non-preferred choice options.

ACC perturbation during strategy re-evaluation increases subsequent acceptance of the non-preferred option

Because re-evaluation of an ongoing strategy of pursuing the preferred option is most likely to occur at the end of a preferred trial (Holroyd and Coles, 2002; Kolling et al., 2014), when the anticipated reward is not delivered, we tested the effect of ACC perturbation when it specifically targeted this trial period (Figure 3A). To avoid perturbing outcome processing itself, we delayed the onset of light delivery by 500 ms relative to reward port entry. Application of light pulses following completion of unrewarded preferred trials resulted in the opposite effect on strategy switching compared with that observed at the commitment point, markedly enhancing the probability of acceptance for the non-preferred option encountered on the subsequent trial (Figures 3B and 3C; ratio of acceptance of non-preferred options following preferred trials with light on versus light off, 1.67 ± 0.21, $p < 0.01$ average of within-animal means, Wilcoxon rank-sum test, n = 5 animals; Figure S3A). Shuffle analysis confirmed that the observed difference in the distribution of accept decisions following light-on and light-off trials did not arise by chance (Figure S3B) and that the effect of perturbation was consistent across individual animals (Figure S3C). Although we did not detect a significant effect of light application on acceptance of the following preferred options (Figures 3B and 3C; ratio of accept decisions for preferred option following light-on versus light-off preferred trials, 1.02 ± 0.01 average of within-animal means, $p > 0.5$, Wilcoxon rank-sum test, n = 5 animals; Figure S3A), we cannot formally exclude the possibility that a "ceiling effect" created by the typically very high baseline rate of acceptance for the preferred option obscured an even greater commitment to accept the subsequently encountered preferred option. However, the clear effect on the non-preferred option argues that ACC perturbation at the time of trial feedback and presumed strategy re-evaluation increases subsequent acceptance of the non-preferred option.

We next again evaluated our conclusion that the observed perturbation effect reflects disruption of a task-related computation within the ACC from the perspective of ACC neural ensemble dynamics. Specifically, we asked whether there was a relationship between ACC neural activity and the likelihood of an animal persisting with the preferred strategy or switching to the alternative after an unrewarded preferred trial. Although the markedly smaller sample size of paired unrewarded preferred trials followed by non-preferred trials made statistical analyses challenging, significant modulation of activity could still be observed at the level of individual cells (Figure 3D, top panel); the majority (72%) of cells displaying significant modulation at the time of trial feedback did not show modulation at the commitment point. The observed modulation was sufficient to train a linear classifier that performed well (Figure 3D, bottom panels; classification accuracy, 68.6% ± 1.0%, n = 3 animals, n = 15 sessions). These findings argue that the ACC actively contributes to computations that permit persistence in the face of transient downturns in reward and raises the question of whether these two opponent computations are segregated by ACC microcircuitry.

The two perturbation effects can be anatomically dissociated via selective access to two ACC subcircuits

Recent work has emphasized the segregation of function in distinct cortical output pathways (Economo et al., 2018; Hirokawa et al., 2019; Li et al., 2020; Vavladeli et al., 2020), prompting

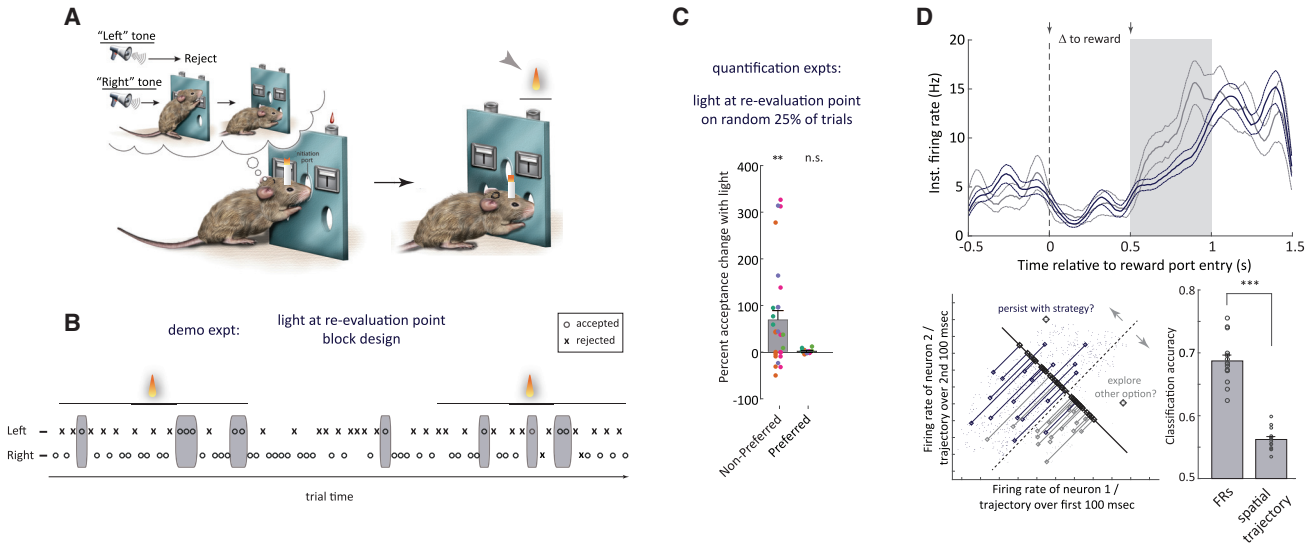

Figure 3. ACC perturbation during strategy re-evaluation increases acceptance of a subsequently encountered non-preferred option

(A) Schematic: light was restricted to feedback/re-evaluation point (arrow).

(B) Example accept/reject behavior. Note higher acceptance of the locally non-preferred option (arrows) during light versus no-light blocks.

(C) Fractional change in accept decisions for non-preferred and preferred options. Colors denote animals. Bars show average effect across within-animal means. n = 5 animals.

(D) Top: example cell modulated by upcoming strategy switching during the re-evaluation period after unrewarded preferred trials. Shown is average activity before trials when the animal persists (blue) and or switches (gray). The box indicates the window following a typical delay to reward delivery ('Δ to reward' in plot) used for decoding. Bottom: schematic of linear classification as in Figure 2. FRs: accuracy for trials in the "persist with strategy" and "switch strategy" groups. Spatial trajectory: accuracy for trials based on associated spatial trajectories. n = 3 animals, n = 15 sessions. Error bars represent SEM. **$p < 0.01$, ***$p < 0.001$. See also Figure S3.

us to next investigate whether the two computations uncovered by our temporally segregated perturbations of the ACC are derived from distinct output pathway-defined ACC microcircuits. Taking advantage of our previously developed retrograde adeno-associated viral vector, AAV2-retro (Tervo et al., 2016), we targeted FLInChR expression to candidate ACC output pathways (STAR Methods), and evaluated the behavioral effect of the corresponding perturbation at the two key time points.

Imbalances in cortico-striatal function have been argued to contribute to behavioral perseveration (Langen et al., 2011), prompting us to hypothesize that computations impinging on the ACC intra-telencephalic (IT) pathway underpin rats' persistence in selectively pursuing the preferred option. We achieved independent access to the IT pathway through its cortico-striatal (CS) collaterals (Figures 4A and S4A; STAR Methods) and examined the behavioral effect of perturbing the IT(CS) cingulate pathway by analyzing the influence of transient photoinhibition on the incidence of accept decisions for non-preferred trials at the two key time points within the task (commitment versus default strategy re-evaluation).

Delivery of light to the ACC of IT-FLInChR rats during the presumed strategy re-evaluation period decreased rats' persistence with the ongoing preferred option; the frequency of accept decisions for the subsequent non-preferred trials increased markedly in all animals under the light-on compared with the light-off condition (Figure 4C; ratio of acceptance of non-preferred option following unrewarded preferred trials with light-on versus light-off, 2.10 ± 0.42, across within-animal means, 2.32 ± 0.43 when one mistargeted animal was removed from the set, $p < 0.01$, Wilcoxon rank-sum test, n = 6[5] animals; Figures S3A and S4A). Shuffle analysis confirmed that that observed difference in the distribution of accept decisions for light-on and light-off trials did not arise by chance (Figure S3B) and that the effect of perturbation was consistent across all properly targeted animals, independent of the specific viral targeting strategy (Figure S3C). Thus, perturbation of the IT(CS)-associated ACC microcircuitry during the default strategy's re-evaluation period mirrored the effect observed when all of the ACC was perturbed during the same period (Figure 3C).

Although perturbation of the ACC IT(CS) pathway alone was sufficient to recapitulate the effect of perturbing the entire ACC during the re-evaluation period, IT(CS) perturbation had no effect at the commitment time point. Indeed, unlike what we observed when the entire ACC was silenced, acceptance of the non-preferred option was not significantly different between light-on versus light-off conditions for IT(CS) perturbation (Figure 4B; ratio of accept decisions for light-on versus light-off trials, 1.03 ± 0.09 across within-animal means, $p > 0.05$, Wilcoxon rank-sum test, n = 4 animals; cf. decrease to 0.77 ± 0.05 for all of the ACC; Figure 1G; Figures S2A and S2B). These observations suggest that either the computation during strategy re-evaluation is more sensitive to partial inactivation of ACC activity than one at the commitment point or that the former computation is selectively implemented within its IT subcircuit. The latter interpretation fits with the observed segregation of units that display activity modulation at re-evaluation or commitment points (see ACC

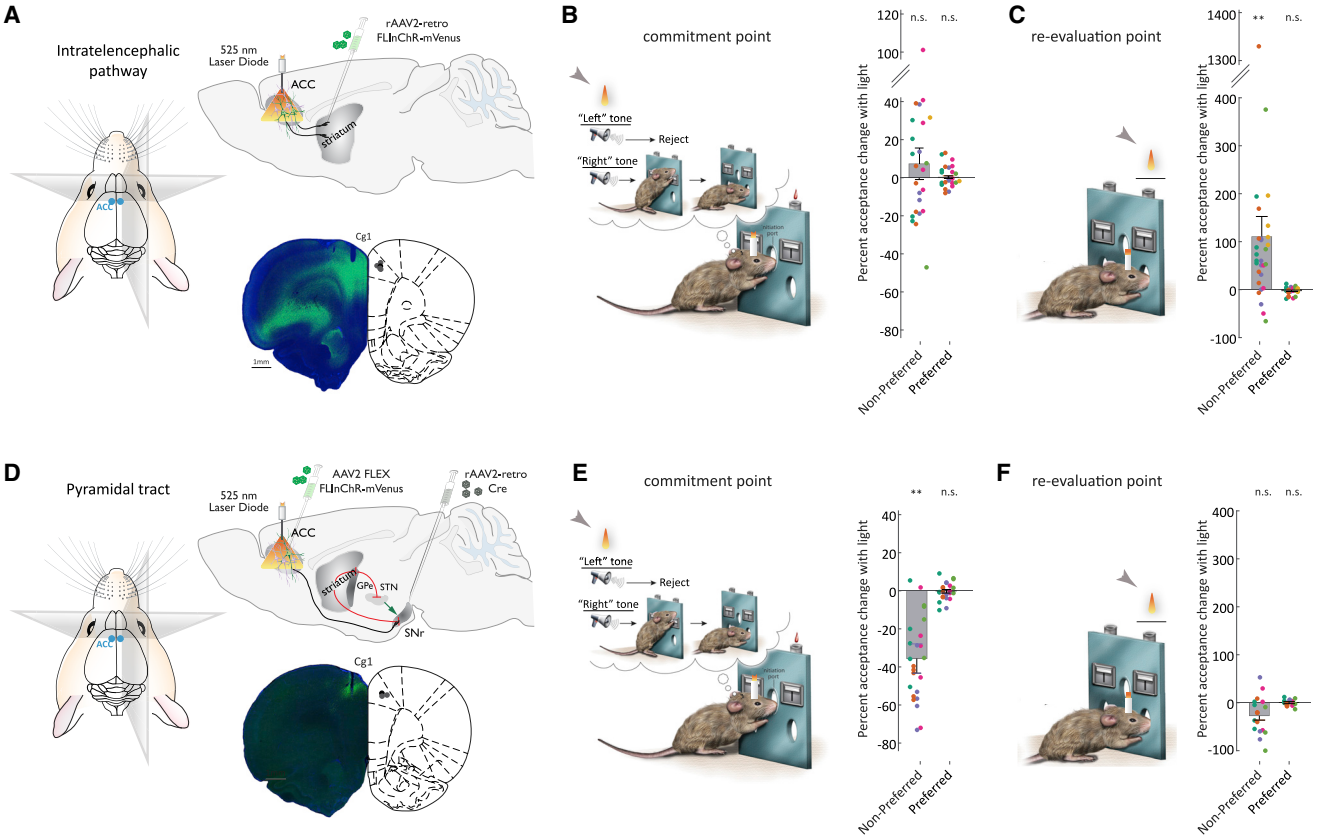

Figure 4. Computations to transiently switch away from the default strategy and to accept a specific alternative are segregated within the ACC microcircuitry

(A) Schematic of the viral approach for access to the IT pathway through its CS branch. Although opsin expression in this example was not restricted to the IT(CS) pathway, specificity for ACC was preserved through localized light delivery.

(B) Fractional change in acceptance of non-preferred and preferred options as a result of IT(CS) perturbation at the commitment point. Note the lack of effect on acceptance of non-preferred options (cf. Figure 1G). n = 6 animals.

(C) Fractional change in acceptance of non-preferred and preferred options following perturbation at the time of default strategy re-evaluation (cf. Figure 3C). n = 6 animals.

(D) Schematic of the viral approach for selective access to the PTs through its cortico-nigral (CN) branch.

(E) Fractional change in acceptance of non-preferred and preferred options as a result of PT(CN) perturbation at the commitment point (cf. Figure 1G). n = 5 animals.

(F) Fractional change in incidence of acceptance of non-preferred and preferred options following perturbation at the time of default strategy re-evaluation. Note the lack of effect on acceptance of non-preferred options (cf. Figure 3C). n = 5 animals. Colors represent individual animals.

Bars show average SEM effect across within-animal means, n = 6 animals. **$p < 0.01$. See also Figures S2–S4.

perturbation during strategy re-evaluation increases subsequent acceptance of the non-preferred option) and with the opposite behavioral effect of perturbing the ACC during those epochs but raises the question of which other ACC subcircuit may implement the latter computation.

Recent work has highlighted the asymmetry of excitatory connectivity in the cortex, with IT neurons impinging on subcortically projecting pyramidal tract (PT) neurons (Brown and Hestrin, 2009; Morishima and Kawaguchi, 2006), providing, among other inputs, feedforward inhibition via local interneurons (Lee et al., 2014; Lu et al., 2017). These anatomical findings suggest that the PT pathway is a candidate downstream strategy switching circuit. We further reasoned that, of the cingulate PT neurons, those projecting to *substantia nigra pars reticulata* (SNr) (Naito and Kita, 1994), the output nucleus of the basal ganglia, are potentially in a privileged position to override ongoing behavioral strategies. The convergent input from the basal ganglia to the SNr and the subsequent divergent outputs from the SNr to the thalamus and superior colliculus position the SNr as a bottleneck that may gate basal ganglion (BG) output. Indeed, the direct pathway striatal input and indirect pathway subthalamic nucleus input converge onto a microcircuit in the *substantia nigra* that is believed to control the gain of this output (Brown et al., 2014; Paxinos, 2004). Thus, BG-dependent motor programs, such as switching away from an ongoing automatic behavior to explore an alternative option (e.g., Hikosaka and Isoda, 2008), could be profoundly affected by a cortical input into the SNr (Doyon and Benali, 2005). The ACC is a natural candidate to provide such input, given multiple proposals that this region contributes to

switching away from ongoing behavioral strategies. We thus sought to selectively target the cingulate PT pathway through its cortico-nigral (CN) branch for perturbation experiments in a separate group of animals.

We achieved independent access to the PT(CN) pathway by combining a bilateral injection of a viral vector that directed conditional expression of FLInChR (AAV-FLEX-FLInChR-mVenus) in the ACC with a bilateral injection into the axonal terminal fields in the medial SNr of a retrograde viral vector that directed expression of Cre recombinase (AAV2-retro Cre) (Figures 4D and S4A). Separately, we verified that bilateral striatal injections of AAV2-retro, which would be expected to target a subset of PT neurons with axon collaterals in the ipsilateral striatum, do not efficiently target CN neurons (Figure S4B; double labeling observed in 1.24% ± 0.79% of cells). In FLInChR-expressing animals, we then analyzed the influence of transient photoinhibition on the incidence of accept decisions for non-preferred trials separately at the commitment point and the point of re-evaluation. Strikingly, we observed a complementary pattern of behavioral impairment compared with that seen with IT-FLInChR animals. Indeed, perturbing the PT(CN) pathway now mirrored the effect on the prevalence of acceptance decisions for the non-preferred choice option observed following ACC perturbation at the commitment point (Figure 4E; ratio of accept decisions for light-on versus light-off trials decreased to 0.64 ± 0.08 across within-animal means, $p < 0.01$, Wilcoxon rank-sum test, $n = 5$ animals, cf. decrease to 0.77 ± 0.05 for all of the ACC; Figure 1G; Figures S2A–S2C) but not at the time of strategy re-evaluation (Figure 4F; ratio of accept decisions for subsequent non-preferred trial following light-on versus light-off preferred trials decreased after PT[CN] perturbation to 0.73 ± 0.09 across within-animal means, $p < 0.01$, Wilcoxon rank-sum test, $n = 5$ animals cf. an increase to 1.67 ± 0.21 for all of the ACC [Figure 3C] and an increase to 2.32 ± 0.43 for the IT[CS] pathway; Figure 4C; Figures S3A and S3B). Whether the unexpected decrease in acceptance of the subsequent non-preferred option (as opposed to a lack of an effect) reflects bleed-through perturbation of the PT(CN) pathway as its activity recovers following the offset of perturbation or some other process, the relevant local computation at this time point, exposed by perturbing all of the ACC or its IT(CS) pathway, is unlikely to take place in this specific subcircuit. Although these combined observations do not preclude a role of the IT and PT pathways in other aspects of adaptive behavior, they indicate a degree of selectivity in persistence and strategy switching. Overall, our findings provide strong causal evidence of the ACC's direct involvement in ongoing strategy arbitration in uncertain environments through a complex opponent interaction of these distinct subcircuits.

DISCUSSION

Dissecting specific contributions of frontal cortices to decision-making has remained challenging because of the experimental trade-off between using a model organism that is behaviorally sophisticated enough for flexible decision-making in complex and volatile environments and one that permits robust circuit dissection with modern molecular tools. Here, we combine a foraging task in rats that requires animals to continuously arbitrate between strategies with pathway-specific optogenetic perturbation to demonstrate that the ACC plays an active role in strategy arbitration. Transient perturbation of ACC activity during the presumed period of default strategy re-evaluation resulted in increased acceptance of subsequently encountered non-preferred options in a shift from the ongoing strategy of exploiting the preferred one. Changing the timing of perturbation to the moment of commitment to an action dictated by a given strategy—uniquely isolated by the iterated nature of option encounter in the task—instead limited the pursuit of the available non-preferred choice option, exposing the ACC's contribution to the ongoing decision. Finally, targeting opsin expression selectively to the IT and PT subcircuits of the ACC permitted anatomical dissociation of the two perturbation effects and provided an internal control for any non-specific effect of viral infection or tissue illumination. These findings complement previous observations that the ACC tracks factors that might contribute to the computation of whether to switch away from the default action (Behrens et al., 2007; Bernacchia et al., 2011; Ito et al., 2003; Kennerley and Wallis, 2009; Kolling et al., 2012, 2016; Matsumoto et al., 2007; Porter et al., 2019; Seo et al., 2014; Shenhav et al., 2016; van Holstein and Floresco, 2020) in demonstrating that this frontal cortical region plays an active role in translating that computation into the selection of behavioral strategies, including during the ongoing decision.

Our finding that perturbing ACC activity at the time of strategy re-evaluation does not lead to the preferred option being rejected demonstrates that the ongoing strategy is not overridden unless the non-preferred option is encountered and argues that the decision of whether to switch away from the ongoing strategy and the actual pursuit of an action dictated by a specific alternative strategy represent two separate computations. Consistent with this interpretation, evidence of two independent, temporally segregated signals, one indexing the growing pressure to pursue non-default choices and one related to the actual making of such choices when the pressure reaches a certain threshold, was found in a recent functional imaging study in human subjects (Kolling et al., 2014). Moreover, our pathway-specific perturbation results suggest the intriguing possibility that there is an anatomical substrate to this functional duality within the ACC microcircuitry; the output of the ACC's PT subcircuit drives the pursuit of alternative learned strategies but is kept clamped through local opponent interaction until evidence grows in favor of deviating from the ongoing strategy. If the representation that guides the pursuit of alternative strategies is indeed embedded in a specific population of cortical projection neurons (the PT tract), then evolution was likely able to tap into one of its principle solutions for achieving a hierarchy of cortical processing: dynamic segregation of discrete projection neuron ensembles via inhibitory control by specialized interneuron subpopulations. Indeed, the functional opponency between the IT(CS) and PT(CN) pathways, both excitatory, would necessarily involve an intervening directional inhibitory pathway.

What might be the nature of the computation subserved by the apparent functional opponency within the ACC microcircuitry? One possibility is that it relates to the general tension between the need to persist with the ongoing strategy in the face of transient downturns and the need to expeditiously detect and adapt

to often unsignaled changes in the environment that render past experience irrelevant for future predictions (Cohen et al., 2007). When the temporal context informs such tension, as in the specific behavioral framework employed in our study, individual differences in resolving it may relate to differences in the estimation of environmental volatility and a growing subjective estimate that a change is impending (Nassar et al., 2010). In this view, the presumably larger-scale network associated with the cingulate's IT pathway would play a key role in computations of the underlying internal temporal expectation. In turn, unprompted switches away from the ongoing strategy would reflect the release of the PT(CN) pathway from inhibitory influence when the subjective estimate of the impending change crosses a certain threshold. In other contexts, the relevant tension may be between making the default choice of persisting with the ongoing strategy and engaging cognitive control processes to determine whether a change in strategy is warranted (Shenhav et al., 2016). Investigating whether control-related computations would similarly engage the cingulate's IT pathway or whether it has a separate anatomical pathway for ultimately releasing the PT from inhibitory influence is an interesting avenue for future studies.

Finally, we note that our observation of a marked modulation of ACC ensemble representation associated with a specific action to which the animal commits, depending on whether that action is part of the ongoing strategy or is being re-evaluated as an alternative, may hint at what is encoded by the PT(CN) pathway. If, in an environment with a richer strategy space, this modulation further scales with the relative prevalence of a specific strategy, then this form of a representation would be empirically equivalent to sampling from a probabilistic distribution over the entertained strategy space (Fiser et al., 2010). Because our results argue that the ACC's PT (SNr) pathway actively guides the exploration of those strategies, demonstration of activity scaling with their local prevalence would suggest that strategies are sampled in accordance with this distribution. Such a finding would lend credence to the idea that the cingulate PT pathway encodes a probabilistic internal model over strategy space.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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AUTHOR CONTRIBUTIONS

D.G.R.T., E.K., and A.Y.K. designed the behavioral training protocol, trained animals to perform the task, and performed optogenetics experiments. M.K. built custom components and wrote software for the electrophysiology setup. M.K., E.K., and M.P. performed the electrophysiology experiments. A.L. built custom components and wrote software for the wireless optogenetics experiments. R.B. generated the viral tools used in this study. M.M., D.G.R.T., E.K., and A.K. analyzed data. D.G.R.T., E.K., and A.K. wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Virus strains		
AAV2/1 pCAMKII- FLInChR-mVenus	this paper	n/a
AAV2-retro pCAG- FLInChR-mVenus	this paper	n/a
AAV2/1 FLEX-CAG-FLInChR-mVenus	this paper	n/a
AAV2-retro pSyn-Cre	this paper	n/a
Experimental models: Organisms/strains		
Long-Evans rat	Charles River Labs	Stock #006
Hardware, software, and algorithms		
Imaging, behavior, electrophysiology, optogenetics analysis code	custom, this paper	n/a
Cerebro	this paper	https://karpova-lab.github.io/cerebro/
MATLAB	MathWorks	n/a
Trodes	SpikeGadgets	https://bitbucket.org/mkarlsso/trodes/src

RESOURCE AVAILABILITY

Lead contact

Requests for data and materials should be addressed to the Lead Contact Alla Karpova (alla@janelia.hhmi.org).

Materials availability

This study did not generate new unique reagents.

Data and code availability

All imaging, electrophysiological, and behavioral data were processed and analyzed using analysis routines custom-written in MATLAB that can be shared upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

All experiments were done in male Long Evans rats (400-500 g). Animals were kept at 85% of their initial body weight before food restriction and maintained on a 12hr light/12hr dark schedule. Experiments were conducted according to National Institutes of Health guidelines for animal research and were approved by the Institutional Animal Care and Use Committee at HHMI's Janelia Farm Research Campus.

18 animals were implanted for optogenetic experiments (6 each targeting all of ACC excitatory neurons, the cortico-striatal population, and the cortico-nigral population). One of the cortico-nigral animals had to be sacrificed prior to collection of any data due to poor recovery from the surgery. All remaining animals were used for perturbation experiments. With one exception, the same set of animals in each of the three experimental group ("all of ACC," "IT-CS" and "PT-CN") was used for perturbation experiments at the point of commitment and the time of strategy re-evaluation. One animal in "all of ACC" group developed an infection around the implant and had to be euthanized early on in the multi-week experiment thus contributing data only to the set of perturbations at the initiation.

1 animal was implanted with a tetrode drive for collection of neural activity during the task. Data from this animal was combined with data from 3 animals used in a previous study (Karlsson et al., 2012).

METHOD DETAILS

Behavioral apparatus and task

All behavior was confined to a box with 23 cm high plastic walls and stainless-steel floors (Island Motion Corp). The floor of the box was 25 cm by 34 cm, and the levers and nose ports were all arranged on one of the 25 cm walls. All lights, nose ports, levers, and reward deliveries were controlled and monitored with a custom-programmed microcontroller, which in turn communicated via USB to a PC running a MATLAB-based control program. Nose port entries were detected with an infrared beam-break detector (IR LED and photodiode pair). The

central initiation port contained one white LED that indicated the option to initiate a new trial. The left and right levers were pneumatically extracted from the wall upon each trial initiation and were retracted after one of the two levers was pressed. Simultaneously upon initiation, one of two sounds was presented across two speakers (located on the two 34cm walls) with equal volume, and they were frequency modulated (1% modulation at 6.67 Hz) around a single base frequency. A 6.5 kHz base frequency signified that the left lever was correct, and a 14 kHz base frequency signified that the right lever was correct. Either tone had an equal chance of being presented upon initiation. Tone presentation was terminated after 2 s or upon the first lever press, whatever came first. All behavior was video recorded at 30 frames/sec using an infrared-sensitive camera. Except after incorrect lever presses, the only visible light source inside the box was from the LED in the initiation port. During error trials, a white LED array in the ceiling of the box was lit, and no trials could be initiated during a timeout period. Liquid rewards (0.1 mL drops of 10% sucrose mixed with black cherry Kool-Aid) were delivered from the reward ports within 0.5 s after port entry with a motorized syringe pump (Harvard Apparatus PHD 2000).

Behavioral training

Food-restricted animals were trained to perform the task with minimal 'shaping'—from the first moment of training, animals were exposed to the full task with four exceptions: 1) animals only needed to press the correct lever once before reward became available, 2) reward probabilities were kept at or above 0.5 for both sides, 3) the timeout period for pressing the wrong lever was kept short (0.5 s) and 4) the time that was allowed to pass between initiating the trial and pressing the lever, as well as between pressing the lever and collecting the reward was 300 s. At first, animals performed the correct sequence of actions rather infrequently and by chance. However, after approximately 1-2 weeks of training, most animals learned the entire task structure, including the presence of the option to reject trials and of unsignaled changes in reward contingencies. After each animal successfully completed two trial blocks in one session, the number of required lever presses was gradually increased to 5, the timeout period was increased to 30 s, and reward probabilities became randomly drawn from a set spanning low and high values (0.2, 0.3, 0.4, 0.5, 0.6, 0.7 and 0.8) at block transitions. Reward probabilities associated with the two sides were selected independently. This design feature made it difficult for the rat to infer the identity of the new and more profitable option simply from the change in reward probability of the preferred option, thus prompting exploratory bouts.

Animals were considered proficient on the task when they preferentially rejected the less profitable side and dynamically updated this rejection policy when reward probabilities changed. 90% of animals became proficient within one month of training. To prevent animals from predicting when reward probability changes would occur, the number of trials within each block was drawn randomly from a Gaussian distribution (mean 250 with standard deviation of 25). Other than the changed reward probabilities, no cues were presented to indicate block transitions.

In vivo virus injections

For localized *in vivo* viral delivery, rats were anaesthetized with isoflurane (~2% by volume in O_2; SurgiVet, Smiths Medical) and a small hole was drilled in the skull above the requisite injection site (see table below). For some injection sites, several injections were made at different depths (see table below). For viral injections, ~250-500 nL of virus-containing solution was slowly injected at each depth into the tissue. Injections were done with a pulled glass pipette (broken and beveled to 25–30 μm (outside diameter); Drummond Scientific, Wiretrol II Capillary Microdispenser) back-filled with mineral oil. A fitted plunger was inserted into the pipette and advanced to displace the contents using a hydraulic manipulator (Narashige, MO-10). Retraction of the plunger was used to load the pipette with virus. The injection pipette was positioned with a Sutter MP-285 manipulator.

The following coordinates were used in this study:

Injection target	Injection coordinates, mm*
ACC (Cg1; Vogt and Paxinos, 2014)	A/P: 2.50; M/L: 0.60; D/V: -2.42, and -2.1
dorsomedial striatum (DMS)	A/P 2.16; M/L 1.7 ; D/V −4.3
SNr	A/P −4.75; M/L ± 2; D/V −8.6, −8.55

* All A/P coordinates are given with respect to Bregma. All D/V coordinates are with respect to the skull.

The following viral constructs were used:

Perturbation target	Viral constructs, injection location	Fiber location
All of ACC	AAV2/1 pCAMKII- FLInChR-mVenus into ACC, titer: 1.9* 10^12 /ml	ACC
Cortico-striatal pathway	AAV2-retro pCAG- FLInChR-mVenus into DMS (4 animals), titer: 2.1*10^12 /ml AAV2-retro pSyn-Cre into DMS, titer: ~2*10^12 /ml AAV2/1 FLEX-CAG-FLInChR-mVenus into ACC (2 animals), titer: 2-3*10^12 /ml	ACC

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Continued

Perturbation target	Viral constructs, injection location	Fiber location
Corticonigral	AAV2-retro pSyn-Cre into SNr, titer: ~2*10^12 /ml AAV2/1 FLEX-CAG-FLInChR-mVenus into ACC titer: 2-3*10^12 /ml	ACC

Wireless optogenetics system

The "Cerebro" optogenetics system consists of an implantable programmable device that is triggered remotely by infrared light. The implanted component comprises two laser diodes (520 nm; Cat No: PL520B1; World Star Tech) coupled to a pair of 200 um core, 0.5 NA hard polymer optic fibers (FP200URT; ThorLabs) cut to protrude from the acrylic base of the implant by approximately 2.5 mm. A photoresistor is used as feedback to maintain a constant light output. Full manufacturing and parts information are provided online (https://karpova-lab.github.io/cerebro/).

To improve light scattering from the optical fibers, two methods were deployed. For some animals, we bonded 106-125 um diameter silver spheres (Cat. No. M-60-0.17, CoSpheric) to the tip of these fiber optics to scatter emitted laser light more uniformly throughout neuropil. For others, we sharpened the fibers using chemical etching approach (Hanks et al., 2015). Specifically, the cable jacket, strengthening fibers, and outer plastic coating (typically white or orange) were fully removed, leaving 1 cm of fiber optic cable and inner plastic coating intact. Then 2 mm of the fiber tip (with final layer of plastic coating still attached) was submerged in 48% hydrofluoric acid topped with mineral oil for 85 min, followed by water for 5 min (submerging 5 mm), and acetone for 2 min (to soften the plastic). The plastic coating was then gently cut with a razor and pulled off with tweezers to reveal a 1 mm sharp-etched fiber tip.

In a subset of animals, the module was implanted during viral injection surgery. In others, the implantation was done as a separate surgery, 3-4 weeks after viral delivery.

Perturbation experiments

To avoid adaptation to perturbation, experiments were run at most every other day, and at times were interspersed with behavioral sessions where no light was delivered.

For perturbation at the time of trial initiation (commitment point), light delivery was done on 25% of all trials. The laser diode was turned on immediately upon central nose port entry, and the delivery of the auditory tone was delayed by 50 msec. If the rat rejected the trial, the laser light waveform was immediately terminated. If the rat accepted the trial, the light waveform was permitted to progress to conclusion (2 s), and then ramped down.

For perturbation during strategy re-evaluation, light delivery was done on 15%–35% of all completed trials; a 2 s waveform was triggered 500 ms after reward port entry (introduction of this delay ensured that processing of the outcome itself was left unperturbed). Since animals typically licked at the reward port even on unrewarded trials, the 2 s perturbation period did not typically extend into the following trial. On rare occasion when animals did proceed to initiate the next trial in less than 2 s, the light waveform was immediately terminated by central port entry.

The two experimental protocols were carried out in different experimental sessions. All perturbations were done bilaterally.

Electrophysiological recordings

Data from 3 out of 4 animals also contributed to previous analyses (Karlsson et al., 2012) and the associated methods were described in detail in that manuscript. The methods for the fourth animals followed a same paradigm except for a different angle of approach and a new recording system. Briefly, a microdrive array containing 20 independently movable tetrodes (O'Keefe and Recce, 1993) was chronically implanted on the head of the animal. Each tetrode was constructed by twisting and fusing together four insulated 13 μm wires (stablohm 800A, California Fine Wire). Each tetrode tip was gold-plated to reduce impedance to 200-300 kΩ at 1 kHz. Within the implant, the tetrodes converged to a circular bundle (1.9 mm diameter), angled either at 20° or at 0° with respect to vertical (pointing toward midline after implantation). Over the two weeks following surgery, the tetrodes were slowly lowered to a depth of 2 mm along the 20° trajectory (for the first three animals), or vertically (for the fourth animal), moving approximately 160 μm/day on average. During this time, animals were re-acclimated to performing the task with the drive. When performance on the task was regained to pre-surgery levels (in terms of motivation and dynamic strategy arbitration behavior), recording sessions began. After each recording session, any tetrodes that did not appear to have any isolatable units were moved down 80 μm.

Each recording session spanned 1.5 to 4 hours, depending on the animal's motivation to perform. Animals were not forced to perform the task and sometimes took breaks (generally around 5 minutes, but sometimes up to 30 minutes). Data from the first three animals were collected with an Nspike data acquisition system (L. Frank, UC San Francisco, and J. MacArthur, Harvard Instrumentation Design Laboratory). Data from the fourth animal was collected using the wireless headstage and datalogger (M. Karlsson, HHMI and SpikeGadgets). An infrared diode array with a large and a small group of diodes was attached to the animal's preamplifier array, and the animal's position in the environment was reconstructed using a semi-automated analysis of digital video of the experiment with custom-written software.

Spike data were sampled at 30 kHz, digitally filtered between 600 Hz and 6 KHz (2 pole Bessel for high and low pass) and threshold crossing events were saved to disk. Continuous local field potential data from all tetrodes was sampled at 1.5 KHz, digitally filtered between 0.5 and 400 Hz, and saved to disk.

Individual units on each tetrode were identified by manually classifying spikes using polygons in two-dimensional views of waveform parameters (Matclust, M.K.). For each channel of a tetrode, peak waveform amplitude and the waveform's projection onto the first two principal components were used for clustering. Autocorrelation analyses were done to exclude units with non-physiological single-unit spike trains. Only units where the entire cluster was visible throughout the recording session were included.

Each animal contributed 7-12 sessions, with 4-56 neurons recorded per session (mean 17.45, median 13). Individual animals contributed 82, 101, 120 and 273 neurons.

Data analysis

All data analysis was performed using custom-written routines in MATLAB.

Identification of 'preferred' and 'non-preferred' trials

Iterated presentation of the two trial types ('left' and 'right') permitted us to make an independent calculation of the running acceptance probability for each behavioral option. pAcceptLeft and pAcceptRight were calculated with a sliding causal half-exponential (decay constant of 0.1), by assigning rejected trials a value of 0 and accepted trails the value of 1. Error trials not included in the calculation.

A trial was deemed 'preferred' ('non-preferred') if the acceptance probability for its cognate choice option was at least (at most) 0.5, and if the concurrent acceptance probability of the other choice option was smaller (greater) by at least 0.3. While all the results were robust to the choice of the latter threshold (the minimum difference in acceptance probability for the two sides), the difference of 0.3 was chosen to balance the desired focus on epochs with a markedly pronounced default strategy (i.e., side preference) and the need to have a sufficient sample size in the 'non-preferred' category.

Periods of preference within one session were pooled and analyzed together.

R.O.C. analysis of features predicting acceptance of non-preferred option

To determine whether, and how well, various behaviorally-relevant features (those present in the task and in the recent behavioral record) predicted the animals' accept decisions for the 'non-preferred' choice option, we used a version of a ROC analysis that that uses the true negative and true positive rates to summarize how well a predictor dichotomizes a dataset, summarized by the Youden's Index, a R.O.C. analysis that uses the true negative and true positive rates to summarize how well a predictor dichotomizes a specific dataset (Youden, 1950). Periods of marked side preference (see above for the difference in acceptance probability deemed sufficient to define preference for one side) from all 160 behavioral sessions were stitched before the ROC analysis.

The behavioral features we examined fell into three main categories: the parameters associated with an individual trial leading up to the choice to be explained (at lags of 1-10 trials), running averages of different types of trials in recent past, and the number of trials since the last occurrence of a particular type of trial. Within the first category, we examined the following features: the type of choice option presented on that trial; whether the trial was accepted/rejected/an error trial; whether the trial was rewarded, whether the trial was both 'preferred' and rewarded, both 'non-preferred' and rewarded. Within the second category, we calculated running averages of: errors, presented choice options, acceptance probability for the 'preferred'/'non-preferred' choice option, and reward probability for 'preferred'/'non-preferred' choice option. In addition, we examined features that reflected the number of trials since the last reward, since the last reward for the 'preferred'/'non-preferred' choice option, since last acceptance, since last acceptance of the 'preferred' / 'non-preferred' choice option since last rejection, since last rejection of the 'preferred'/ 'non-preferred' choice option, since the last presentation of the 'preferred'/'non-preferred' choice option presentation, since the beginning of the estimated start of a stable side preference, and since the beginning of a reward block. Finally, we looked at the experienced reward rates within each block, and the absolute difference in rejection of the 'preferred' and 'non-preferred' choice options. Each feature was examined within the space of all trials, and, when appropriate separately, within the subspace of trails of the relevant type.

Our final analysis (Figure 1C) focused on estimating the Youden's Index for three features that performed significantly better than others at capturing the decisions to accept the 'non-preferred' choice option: the number of trials (in the space of all trials) since the last rewarded 'preferred' trial, local probability of accepting the 'non-preferred' choice option (in the subspace of the 'non-preferred' trial type), and the number of consecutive encounters with the 'non-preferred' choice option (equivalent to the number of trials since the last 'preferred' choice option). For comparison, Youden's Index was also calculated for one of the lesser-predictive features, the local error rate.

Effect of environmental volatility on acceptance of the 'non-preferred' choice option

The capacity of local features to predict animals' choices suggests that, in principle, a purely local strategy have been employed. To determine if, instead, higher-order task variables also influence animals' acceptance of the 'non-preferred' choice option, we examined whether the prevalence of such decisions changed with the number of trials into a block of stable reward probabilities. To uncover the influence of such more global features in face of a strong contribution from the local ones, we evaluated how the prevalence of accepting the 'non-preferred' option changed as a joint function of the relevant global feature – time since the block start – and of

individual local features identified above. Specifically, we plotted the prevalence of accepting the 'non-preferred' choice option as a function of the block depth (in 30-trial bins; normalized by the prevalence in bin 1) and of the different parameterizations of the local features (calculated for the 10-trial window). Similarly to the ROC analysis above, this analysis was performed on the entire behavioral dataset as a whole, with periods of marked behavioral preference stitched together across all sessions.

Analysis of optogenetic perturbation experiments

We calculated the ratio of acceptance probability between 'light-on' and 'light-off' conditions within each session for the trial type of interest (Figures 1, 3, and 4). Statistics on the observed ratios were done across within-animal means. To guard against battery run-down in the optogenetic module, the upper limit on the number of trials used in the analysis was set at 1200 for the sessions with the perturbation during the commitment point, and to 1500 for sessions with perturbation during strategy re-evaluation point.

Shuffle analysis

To eliminate the possibility that the difference in the distribution of accept decisions for 'light-on' and 'light-off' trials occurred by chance, we repeated our analysis after randomly shuffling the 'light-on' and 'light-off' labels within individual sessions. If our observations arose by chance, such a manipulation would be expected to frequently produce values of mean fractional change in acceptance on the order of what we observed experimentally. Within each shuffle run, 'light' and 'non-light' labels would be independently shuffled for each individual experimental session for an individual condition (e.g., "all of ACC" @ commitment point). For perturbation sessions, during which the light stimulation occurred on initiation trials, any trial in the synthetic shuffle sessions could be designated as a light stimulation trial. For sessions, in which the light stimulation occurred during strategy re-evaluation, only trials that were not errors and were accepted could acquire the 'light' label to ensure that the synthetic shuffle sessions randomly distributed synthetic light stimulation on these correct, accepted trials. The change in accept decisions for 'non-preferred' and 'preferred' option as a result of this synthetic 'light' delivery was then computed across that run of generating synthetic session in the same manner as was done for the experimental data. This procedure was repeated 10,000 times, and the "null distribution" of such synthetic mean percent changes was then constructed. The percentile value of experimentally observed mean percent change inside this 'null' shuffle distribution was calculated as the measure of significance of the result. When shuffle effects were evaluated for individual animals, the null distribution was constructed separately in each case, only using data for that animal.

Decoding analysis

The most direct logical way to evaluate whether neural activity in the ACC shows the relevant contextual modulation is to compare the neural responses between the concurrent 'preferred' and 'non-preferred' choice options. For example, for the analysis at the commitment time point, one would intuitively want to compare the neural responses when the animal commits to either the locally preferred or to the concurrently non-preferred choice option. However, at any given time, the specific actions associated with accepting the preferred and the non-preferred choice option are distinct, involving moving through space to two different locations, and ACC displays pronounced spatial and trajectory encoding. Given that the animal's body position (starting with an immediate slight tilt of the head one way or another as he hears the tone) and movement through space are different between accepting the 'left' tone versus accepting the 'right' tone, such an analysis is certainly bound to demonstrate a difference in encoding, but that result would be inescapably confounded.

A more rigorous way to test whether the activity in ACC displays a relevant contextual modulation is thus to ask whether ACC activity associated with the same action (e.g., 'accept right side') displays modulation according to the preferred/non-preferred context. To accomplish this, we trained, for each session, a linear classifier to predict, on the basis of ensemble activity, whether a particular trial was 'preferred' (part of the 'default' strategy) or non-preferred (part of the 'resurgent' strategy). Because of a strong spatial component to neural activity in ACC, we trained a separate classifier for left trials and right trials. If there was a trial group—left-bound or right-bound—that had less than ten trials of a trial type—preferred or non-preferred—that accuracy was not calculated. Otherwise, for each of 200 iterations, we used a random 80% of 'default' and 80% '' trials to train the classifier. All cells that had a non-zero firing rate were included in the analysis; the firing rate for each was calculated over the entire 500 msec decoding window. We performed linear discriminant analysis on the training dataset; once the direction normal to the discriminant hyperplane was determined, we selected the best classification threshold from the following three threshold candidates:

- the midpoint between the group centroids
- the intersection point of the probability density functions for the two groups
- the intersection point of the cumulative density functions for the two groups.

Classification accuracy was estimated on the remaining 20% of the dataset, after ensuring that the sample size in each category was the same. For 'left-side option' and 'right-side option' classifiers, we averaged the accuracy from 200 cross-validation runs for each session. Average accuracy for the better performing of the two classifiers was reporter for each session. Statistics on the estimated classification accuracy was done across sessions.

To control for possible contribution to the difference in ensemble activity between the two conditions from the reward context (high versus low reward), we focused on sessions where reward probability changes were limited to reversals, i.e., where the value of

reward probability for each option ('Left' and 'Right') was either 'high' (typically, 0.5 or 0.6) and 'low' (typically, 0.25 or 0.3). Because the definition of 'preferred' versus 'non-preferred' trials is based on the animal's behavior rather than experimentally defined reward contingencies, and because animals often persisted with preferring one option for some time past the block switch, 'preferred' and 'non-preferred' trial groups had a mixture of trials that fell within 'high' and 'low' reward contingency for that choice option. This mixing gave us an opportunity to repeat FR-based classification for trials from the preferred and non-preferred categories that were matched for local reward rates.

To control for possible contribution to the difference in ensemble activity between the two conditions from the way, in which an animal was executing the trials, we also estimated classification accuracy for a linear classifier trained on spatial trajectories rather than ensemble firing rates. The spatial trajectory for each trail was represented either as the mean X and Y positions from 100 msec bins or as the first five principal components of the raw video capture of the animal's X and Y positions.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were conducted using Wilcoxon rank-sum test. The n value, the mean values ± SEM for each dataset, and statistically significant effects are reported in figure legends and are summarized in the table below. The significance threshold was held at $\alpha = 0.05$. The individual p values can be found in Table S1.